

$$\begin{aligned}
& y_u^{i1i2} + \frac{1}{16\pi^2} \left(\frac{1}{48} (6 y_u^{pi2} (\overline{y_d^{pr}} y_d^{i1r} + 3 \overline{y_u^{pr}} y_u^{i1r}) + y_u^{i1i2} (11 g_1^2 + 9 g_2^2 + 96 g_3^2)) - \frac{1}{36} g_1^2 y_u^{i1i2} \right. \\
& \quad LF_{1,1,0} [m_1, m_q^{i1}] + \frac{1}{72} g_1^2 y_u^{i1i2} LF_{2,1,-1} [m_1, m_q^{i1}] - \frac{4}{9} g_1^2 y_u^{i1i2} LF_{1,1,0} [m_1, m_u^{i2}] + \frac{2}{9} g_1^2 \\
& \quad y_u^{i1i2} LF_{2,1,-1} [m_1, m_u^{i2}] - \frac{3}{4} g_2^2 y_u^{i1i2} LF_{1,1,0} [m_2, m_q^{i1}] + \frac{3}{8} g_2^2 y_u^{i1i2} LF_{2,1,-1} [m_2, m_q^{i1}] - \\
& \quad \frac{4}{3} g_3^2 y_u^{i1i2} LF_{1,1,0} [m_3, m_q^{i1}] + \frac{2}{3} g_3^2 y_u^{i1i2} LF_{2,1,-1} [m_3, m_q^{i1}] - \frac{4}{3} g_3^2 y_u^{i1i2} LF_{1,1,0} [m_3, m_u^{i2}] + \\
& \quad \frac{2}{3} g_3^2 y_u^{i1i2} LF_{2,1,-1} [m_3, m_u^{i2}] - \frac{1}{2} \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{1,1,0} [m_d^r, \tilde{\mu}] + \\
& \quad \frac{1}{2} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} y_u^{i1i2} LF_{2,1,0} [m_l^p, m_e^r] - \frac{1}{2} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} y_u^{i1i2} LF_{3,1,-1} [m_l^p, m_e^r] + \\
& \quad \tilde{\mu} y_e^{pr} y_u^{i1i2} (\overline{C_{\phi2\phi1}} \overline{a_e^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_e^{pr}}) LF_{3,1,0} [m_l^p, m_e^r] - \\
& \quad 3 \tilde{\mu} y_e^{pr} y_u^{i1i2} (\overline{C_{\phi2\phi1}} \overline{a_e^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_e^{pr}}) LF_{4,1,-1} [m_l^p, m_e^r] + \\
& \quad 2 \tilde{\mu} y_e^{pr} y_u^{i1i2} (\overline{C_{\phi2\phi1}} \overline{a_e^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_e^{pr}}) LF_{5,1,-2} [m_l^p, m_e^r] + \\
& \quad \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} y_u^{i1i2} LF_{2,1,0} [m_q^p, m_d^r] - \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} y_u^{i1i2} LF_{3,1,-1} [m_q^p, m_d^r] + \\
& \quad 3 \tilde{\mu} y_d^{pr} y_u^{i1i2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{3,1,0} [m_q^p, m_d^r] - \\
& \quad 9 \tilde{\mu} y_d^{pr} y_u^{i1i2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{4,1,-1} [m_q^p, m_d^r] + \\
& \quad 6 \tilde{\mu} y_d^{pr} y_u^{i1i2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{5,1,-2} [m_q^p, m_d^r] - \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{1,1,0} [m_q^p, \tilde{\mu}] + \\
& \quad \frac{3}{2} y_u^{i1i2} \overline{a_u^{pr}} a_u^{pr} LF_{2,1,0} [m_u^r, m_q^p] - \frac{3}{2} y_u^{i1i2} \overline{a_u^{pr}} a_u^{pr} LF_{3,1,-1} [m_u^r, m_q^p] + \\
& \quad 3 y_u^{i1i2} \overline{a_u^{pr}} (C_{\phi1^2} a_u^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr}) LF_{3,1,0} [m_u^r, m_q^p] - \\
& \quad 9 y_u^{i1i2} \overline{a_u^{pr}} (C_{\phi1^2} a_u^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr}) LF_{4,1,-1} [m_u^r, m_q^p] + \\
& \quad 6 y_u^{i1i2} \overline{a_u^{pr}} (C_{\phi1^2} a_u^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr}) LF_{5,1,-2} [m_u^r, m_q^p] - \frac{1}{2} \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{1,1,0} [m_u^r, \tilde{\mu}] - \\
& \quad \frac{3}{4} g_1^2 y_u^{i1i2} LF_{2,1,-1} [\tilde{\mu}, m_1] + \frac{1}{2} g_1^2 y_u^{i1i2} LF_{3,1,-2} [\tilde{\mu}, m_1] - \\
& \quad 2 C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{3,1,-1} [\tilde{\mu}, m_1] + m_1 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{3,1,0} [\tilde{\mu}, m_1] + \\
& \quad 4 C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{4,1,-2} [\tilde{\mu}, m_1] - 3 m_1 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{4,1,-1} [\tilde{\mu}, m_1] - \\
& \quad 2 C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{5,1,-3} [\tilde{\mu}, m_1] + 2 m_1 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{5,1,-2} [\tilde{\mu}, m_1] - \\
& \quad \frac{9}{4} g_2^2 y_u^{i1i2} LF_{2,1,-1} [\tilde{\mu}, m_2] + \frac{3}{2} g_2^2 y_u^{i1i2} LF_{3,1,-2} [\tilde{\mu}, m_2] - \\
& \quad 6 C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{3,1,-1} [\tilde{\mu}, m_2] + 3 m_2 \overline{C_{\phi2\phi1}} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{3,1,0} [\tilde{\mu}, m_2] + \\
& \quad 12 C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{4,1,-2} [\tilde{\mu}, m_2] - 9 m_2 \overline{C_{\phi2\phi1}} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{4,1,-1} [\tilde{\mu}, m_2] - \\
& \quad 6 C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{5,1,-3} [\tilde{\mu}, m_2] + 6 m_2 \overline{C_{\phi2\phi1}} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{5,1,-2} [\tilde{\mu}, m_2] + \\
& \quad \frac{1}{4} \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{2,1,-1} [\tilde{\mu}, m_d^r] + \frac{1}{2} \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{2,1,-1} [\tilde{\mu}, m_q^p] + \\
& \quad \frac{1}{4} \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{2,1,-1} [\tilde{\mu}, m_u^r] + \frac{2}{9} m_1 g_1^2 a_u^{i1i2} LF_{1,1,1,0} [m_1, m_q^{i1}, m_u^{i2}] - \\
& \quad \frac{1}{18} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{2,2,1,-1} [m_1, m_q^{i1}, m_u^{i2}] - \\
& \quad \frac{1}{6} g_1^2 y_u^{i1i2} LF_{1,1,1,-1} [m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{18} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) \\
& \quad LF_{2,2,1,-1} [m_1, m_u^{i2}, m_q^{i1}] + \frac{2}{3} g_1^2 y_u^{i1i2} LF_{1,1,1,-1} [m_1, m_u^{i2}, \tilde{\mu}] - \\
& \quad \frac{1}{12} C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_q^{i1}] + \frac{1}{12} m_1 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i1}] + \\
& \quad \frac{1}{3} C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_u^{i2}] - \frac{1}{3} m_1 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i2}] + \\
& \quad \frac{3}{2} g_2^2 y_u^{i1i2} LF_{1,1,1,-1} [m_2, m_q^{i1}, \tilde{\mu}] + \frac{3}{4} C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{2,2,1,-2} [m_2, \tilde{\mu}, m_q^{i1}] - \\
& \quad \frac{3}{4} m_2 \overline{C_{\phi2\phi1}} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i1}] + \frac{8}{3} m_3 g_3^2 a_u^{i1i2} LF_{1,1,1,0} [m_3, m_q^{i1}, m_u^{i2}] - \\
& \quad \frac{2}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{2,2,1,-1} [m_3, m_q^{i1}, m_u^{i2}] - \\
& \quad \frac{2}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{2,2,1,-1} [m_3, m_u^{i2}, m_q^{i1}] + \\
& \quad \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{1,1,1,0} [m_d^r, m_q^p, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) \\
& \quad LF_{2,1,1,0} [m_d^r, m_q^p, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{3,1,1,-1} [m_d^r, m_q^p, \tilde{\mu}] - \\
& \quad \frac{1}{4} \tilde{\mu} y_d^{i1r} y_u^{pi2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{2,2,1,-1} [m_d^r, \tilde{\mu}, m_q^p] + \\
& \quad \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{2,1,1,0} [m_q^p, m_d^r, \tilde{\mu}] - \\
& \quad \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{3,1,1,-1} [m_q^p, m_d^r, \tilde{\mu}] - \\
& \quad \frac{1}{4} \tilde{\mu} y_d^{i1r} y_u^{pi2} (\overline{C_{\phi2\phi1}} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_d^r] + \\
& \quad \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0} [m_q^{i1}, m_1, m_u^{i2}] - \\
& \quad \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1} [m_q^{i1}, m_1, m_u^{i2}] + \\
& \quad \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0} [m_q^{i1}, m_3, m_u^{i2}] - \\
& \quad \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1} [m_q^{i1}, m_3, m_u^{i2}] + \\
& \quad \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0} [m_u^{i2}, m_1, m_q^{i1}] - \\
& \quad \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1} [m_u^{i2}, m_1, m_q^{i1}] + \\
& \quad \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0} [m_u^{i2}, m_3, m_q^{i1}] - \\
& \quad \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1} [m_u^{i2}, m_3, m_q^{i1}] \Big)
\end{aligned}$$